4.2 Types of programming language, translators and integrated development environments (IDEs) continued

Candidates should be able to:	Notes and guidance
4 Explain the advantages and disadvantages of a compiler and an interpreter	• Including an understanding that an interpreter is mostly used when developing a program and a compiler is used to translate the final program
5 Explain the role of an IDE in writing program code and the common functions IDEs provide	• Including: – code editors – run-time environment – translators – error diagnostics – auto-completion – auto-correction – prettyprint

5 The internet and its uses

5.1 The internet and the world wide web

Candidates should be able to:	Notes and guidance
1 Understand the difference between the internet and the world wide web	• The internet is the infrastructure • The world wide web is the collection of websites and web pages accessed using the internet
2 Understand what is meant by a uniform resource locator (URL)	• A URL is a text-based address for a web page; it can contain the protocol, the domain name and the web page/file name
3 Describe the purpose and operation of hypertext transfer protocol (HTTP) and hypertext transfer protocol secure (HTTPS)	
4 Explain the purpose and functions of a web browser	• The main purpose of a web browser is to render hypertext markup language (HTML) and display web pages • Functions include: – storing bookmarks and favourites – recording user history – allowing use of multiple tabs – storing cookies – providing navigation tools – providing an address bar

5.1 The internet and the world wide web continued

Candidates should be able to:	Notes and guidance
5 Describe how web pages are located, retrieved and displayed on a device when a user enters a URL	• Including the role of: – the web browser – IP addresses – the domain name server (DNS) – the web server – HTML
6 Explain what is meant by cookies and how they are used, including session cookies and persistent cookies	• Cookies are used for functions, including: – saving personal details – tracking user preferences – holding items in an online shopping cart – storing login details

5.2 Digital currency

Candidates should be able to:	Notes and guidance
1 Understand the concept of a digital currency and how digital currencies are used	• A digital currency is one that only exists electronically
2 Understand the process of blockchain and how it is used to track digital currency transactions	• Blockchain, in its basic form, is a digital ledger, that is a time-stamped series of records that cannot be altered

5.3 Cyber security

Candidates should be able to:	Notes and guidance
1 Describe the processes involved in, and the aim of carrying out, a range of cyber security threats	• Including: – brute-force attack – data interception – distributed denial of service (DDoS) attack – hacking – malware (virus, worm, Trojan horse, spyware, adware, ransomware) – pharming – phishing – social engineering

5.3 Cyber security continued

Candidates should be able to:	Notes and guidance
2 Explain how a range of solutions are used to help keep data safe from security threats	• Including: – access levels – anti-malware, including anti-virus and anti-spyware – authentication (username and password, biometrics, two-step verification) – automating software updates – checking the spelling and tone of communications – checking the URL attached to a link – firewalls – privacy settings – proxy-servers – secure socket layer (SSL) security protocol

6 Automated and emerging technologies

6.1 Automated systems

Candidates should be able to:	Notes and guidance
1 Describe how sensors, microprocessors and actuators can be used in collaboration to create automated systems	
2 Describe the advantages and disadvantages of an automated system used for a given scenario	• Including scenarios from: – industry – transport – agriculture – weather – gaming – lighting – science